HK IMO Training 2019-2020
Problem Set 13 Suggested Solutions

Problem 49. A hotel has 100 rooms with capacities for $101, 102, \ldots, 200$ people, respectively. These rooms are occupied by n people in total. Banach arrives, and Hilbert, the owner, wants to provide him with a personal room. For that purpose, Hilbert chooses two rooms, A and B, and moves all the people from room A to room B without exceeding its capacity. Determine the largest n for which Hilbert can be sure he is able to reach the goal whatever the initial distribution of people is.

Solution. The answer is 8824.

Let $a_{101}, a_{102}, \ldots, a_{200}$ be the number of people in the rooms with capacities $101, 102, \ldots, 200$ respectively. If $a_i = 76$ for $101 \leqslant i \leqslant 150$ and $a_i = i - 75$ for $i \geqslant 151$, we claim that it is impossible to reach the goal. This means we need to prove $a_i + a_j > j$ for any $101 \leqslant i < j \leqslant 200$.

Indeed, for $101 \leqslant i < j \leqslant 150$, we have $a_i + a_j = 152 > j$. For $101 \leqslant i \leqslant 150 < j \leqslant 200$, we have $a_i + a_j = j + 1 > j$. For $150 < i < j \leqslant 200$, we have $a_i + a_j = i + j - 150 > j$. So this gives an example of $76 \times 50 + \sum_{i=151}^{200} (i - 75) = 8825$ people such that the goal cannot be reached.

Now, suppose $a_{101} + a_{102} + \cdots + a_{200} \leqslant 8824$. Noting that $8824 = (152 + 153 + \cdots + 201) - 1$, one of the following inequalities must hold: $a_i + a_{301-i} \leqslant 301 - i$ for $i = 101, 102, \ldots, 150$. Suppose the inequality $a_k + a_{301-k} \leqslant 301 - k$ holds for some k. Then we can move all people in the kth room to the $(301 - k)$th room to reach the goal. $\square$

Problem 50. Suppose that a competition has 300 participants. For any 3 participants there are 2 that are not friends. Euler enumerates the participants in some order and denotes by x_i the number of friends of the ith participant. If

$$\{x_1, x_2, \ldots, x_{299}, x_{300}\} = \{1, 2, \ldots, N - 1, N\},$$

find the largest possible value of N.

Solution. The answer is 200.

We use terminologies in graph theory. Firstly, we shall prove that if A and B are adjacent, then the sum of degrees of A and B does not exceed 300. Indeed, for each of the remaining 298 vertices, it can be adjacent to at most one of A and B by the given condition. Together with the edge AB, which is counted twice in the sum of degrees, the sum of degrees of A and B is at most $298 + 2 = 300$.

Suppose on the contrary that a vertex X has degree at least 201. Then there exist vertices $Y_{100}, Y_{101}, \ldots, Y_{200}$ with degrees $100, 101, \ldots, 200$ respectively. By the above claim, X is not adjacent to each Y_i. But then the degree of X can be at most $300 - 100 = 200$, which is a contradiction. Therefore, $N \leqslant 200$.

Here is an example for $N = 200$. Label the vertices $A_1, A_2, \ldots, A_{200}, B_1, B_2, \ldots, B_{100}$. For each $i = 1, 2, \ldots, 100$, draw an edge bewteen B_i and each of $A_i, A_{i+1}, \ldots, A_{200}$. Then the resultant graph is a bipartite graph and so it cannot contain any triangle. Also, the degrees of $A_1, A_2, \ldots, A_{100}$ are $1, 2, \ldots, 100$ respectively, while the degrees of $B_1, B_2, \ldots, B_{100}$ are $200, 199, \ldots, 101$ respectively. $\square$

Problem 51. Consider an equilateral triangular array of numbers consisting of 2019 rows with k numbers in the kth row, $k = 1, 2, \ldots, 2019$. Except for the numbers in the bottom row, each number is equal to the sum of two numbers immediately below it. How many ways can one assign 0 or 1 to each of the numbers $a_0, a_1, \ldots, a_{2018}$ (from left to right) in the bottom row such that the number S on the top is divisible by 1009?

Solution. The answer is 2^{2016}.

First, by induction, one can show that

$$S = \binom{2018}{0} a_0 + \binom{2018}{1} a_1 + \cdots + \binom{2018}{2018} a_{2018}.$$

Let $p = 1009$, which is a prime number. We now claim that $\binom{2p}{r}$ is divisible by p if $p \nmid r$, and $\binom{2p}{p} \equiv 2 \pmod{p}$. (Indeed, these follow directly from Lucas' theorem.)

If $p \nmid r$, since $\binom{2p}{r} = \binom{2p}{2p-r}$, we may assume $1 \leqslant r \leqslant p-1$. We have

$$\binom{2p}{r} = \frac{(2p)(2p-1)\cdots(2p-r+1)}{r!}.$$

The numerator is divisible by p while the denominator does not contain the factor p. Therefore, $p \mid \binom{2p}{r}$.

Next, we have

$$\binom{2p}{p} = \frac{(2p)(2p-1)\cdots(p+1)}{p!} = \frac{2(2p-1)(2p-2)\cdots(p+1)}{(p-1)!}.$$

Since $2p-1, 2p-2, \ldots, p+1$ are $p-1$ consecutive numbers not divisible by p, they form a reduced set of residues modulo p. Thus, the numerator is congruent to $2 \cdot (p-1)!$ modulo p. This implies $\binom{2p}{p} \equiv 2 \pmod{p}$.

Using these results, S is divisible by 1009 if and only if $a_0 + 2a_{1009} + a_{2018}$ is divisible by 1009. This holds if and only if $a_0 = a_{1009} = a_{2018} = 0$. There is no restriction on all other terms. Therefore, the number of ways is 2^{2016}. $\square$

Problem 52. A sequence $\{a_n\}_{n \geqslant 1}$ of real numbers is defined as follows:

$$a_1 = 2, \quad a_{n+1} = \frac{a_n^2 + 1}{2}, \text{ for } n \geqslant 1.$$

Prove that

$$\sum_{j=1}^{N} \frac{1}{a_j + 1} < 1$$

for every $N = 1, 2, \ldots$.

Solution. For any $j \geqslant 1$, we have

$$a_{j+1} - 1 = \frac{a_j^2 - 1}{2} = \frac{(a_j - 1)(a_j + 1)}{2}.$$

It is easy to show by induction that $a_j > 1$ for all j. So this can be rewritten as

$$\frac{1}{a_j + 1} = \frac{1}{2}\left(\frac{a_j - 1}{a_{j+1} - 1}\right). \tag{1}$$

Next, for any $j \geqslant 1$, we also have

$$a_{j+1} - a_j = \frac{a_j^2 + 1}{2} - a_j = \frac{(a_j - 1)^2}{2}. \tag{2}$$

Combining (1) and (2), we get

$$\begin{aligned}
\sum_{j=1}^{N} \frac{1}{a_j + 1} &= \frac{1}{2} \sum_{j=1}^{N} \frac{a_j - 1}{a_{j+1} - 1} \\
&= \frac{1}{2} \sum_{j=1}^{N} \frac{(a_j - 1)^2}{(a_{j+1} - 1)(a_j - 1)} \\
&= \sum_{j=1}^{N} \frac{a_{j+1} - a_j}{(a_{j+1} - 1)(a_j - 1)} \\
&= \sum_{j=1}^{N} \left(\frac{1}{a_j - 1} - \frac{1}{a_{j+1} - 1} \right) \\
&= \frac{1}{a_1 - 1} - \frac{1}{a_{N+1} - 1} \\
&< \frac{1}{a_1 - 1} = 1.
\end{aligned}$$

for every $N = 1, 2, \ldots$. $\square$